



GRADE 8
STEM OLYMPIAD
National Level Examination (NLE) 2026

SUBJECT CODE			
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Total Questions: 50

Total Marks: 50

Time: 75 Minutes

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED TO DO SO

- **Carefully Check Your Name, Class, Roll Number, And Other Required Details On The OMR Answer Sheet.**
- **Do Not Forget To Sign The OMR Sheet. Also Write Your Roll No. On The Question Paper Booklet.**
- **Use Only Black/Blue Ball Pen And HB Pencil To Fill The OMR Answer Sheet.**
- **All Questions Are Compulsory And Carry One Mark Each.**
- **Section A: Mathematics & Quantitative Reasoning (15 Questions); Section B: Science & Scientific Reasoning (15 Questions); Section C: Engineering, Technology & Data (10 Questions); Section D: STEM Olympiad Challenge (10 Questions).**
- **Only OMR Answer Sheets Will Be Evaluated.**
- **No Negative Marking For Incorrect Answers.**
- **You May Use The Space Provided On The Last Page For Rough Work (If Needed).**
- **Digital Devices Like Calculators, Smart Watches, Or Mobile Phones Are Not Allowed.**
- **After The Exam, Submit Both The OMR Sheet And The Question Paper To The Invigilator.**

SECTION A – MATHEMATICS & QUANTITATIVE REASONING

Q1. What is $(-3/5) \div (9/10)$?

- A. $-2/3$
- B. $-3/2$
- C. $2/3$
- D. $3/2$

Q2. Solve: $5x - 9 = 3x + 17$.

- A. $x=11$
- B. $x=12$
- C. $x=13$
- D. $x=14$

Q3. What is $\sqrt{2025}$?

- A. 35
- B. 40
- C. 45
- D. 50

Q4. What is the cube root of 3375?

- A. 10
- B. 12
- C. 15
- D. 25

Q5. Simplify: $2^5 \times 4^3 \div 2^4$.

- A. 2^5
- B. 2^6
- C. 2^7
- D. 2^8

Q6. A price increases from Rs 800 to Rs 920. Percentage increase is:

- A. 12%
- B. 15%
- C. 18%
- D. 20%

Q7. A cylinder has radius 3 cm and height 10 cm. Using $\pi=3.14$, its volume is about:

- A. 94.2 cm^3
- B. 188.4 cm^3
- C. 282.6 cm^3
- D. 314 cm^3

Q8. A rectangle is enlarged so length and width both become 1.5 times. Its area becomes:

- A. 1.5 times
- B. 2 times
- C. 2.25 times
- D. 3 times

Q9. The mean of 5 numbers is 18. If four are 12, 16, 20 and 22, the fifth is:

- A. 18
- B. 20
- C. 22
- D. 24

Q10. Probability of drawing an ace from a standard 52-card deck is:

- A. $1/52$
- B. $1/26$
- C. $1/13$
- D. $4/13$

Q11. If 8 pumps fill a tank in 6 h, 12 identical pumps fill it in:

- A. 3 h
- B. 4 h
- C. 6 h
- D. 9 h

Q12. The diagonal of a rectangle divides it into two triangles of:

- A. equal area
- B. always different area
- C. zero area
- D. unrelated area

Q13. A number is written as 5.6×10^4 . Its ordinary form is:

- A. 5,600
- B. 56,000
- C. 560,000
- D. 0.00056

Q14. Which expression is a perfect square?

- A. $2^3 \times 3^2$
- B. $2^4 \times 3^2$
- C. $2^5 \times 3$
- D. 2×3^3

Q15. A graph rises 12 units when x rises 3 units. Its constant rate of change is:

- A. 3
- B. 4
- C. 9
- D. 36

SECTION B – SCIENCE & SCIENTIFIC REASONING

Q16. Microorganisms that convert milk into curd are mainly:

- A. Lactobacillus bacteria
- B. viruses
- C. algae only
- D. protozoa only

Q17. Why is water generally used to extinguish a fire?

- A. It increase the temperature
- B. It lowers the temperature of the burning substance
- C. It Supplies Oxygen
- D. It acts as a fuel

Q18. A Farmer wants to store grains for several?

- A. Adding water to grains
- B. Drying the grains exposed properly before storage
- C. Leaving grains exposed to rain
- D. Keeping wet grains in a closed container

Q19. Suppose the world continues to consume fossil fuels at the present rate without developing alternatives. Which problems is not likely to become serious in the long term?

- A. Fossil fuel reserve may become depleted
- B. Oxygen will completely disappears
- C. More fossil fuel will form quickly
- D. Petroleum will become renewable

Q20. Friction can be reduced in machine parts by:

- A. lubrication
- B. roughening surfaces
- C. adding sand
- D. increasing load

Q21. A School installs curtains and carpets in the its auditorium to reduce unwanted sound . How do these materials help?

- A. They increase sound reflection
- B. They absorb some sound and reduce reflections
- C. They increase the frequency of sound
- D. They produce louder sound

Q22. The basic structural and functional unit of life is the:

- A. cell
- B. tissue
- C. organ
- D. organ system

Q23. In humans, fertilisation usually occurs in the:

- A. oviduct
- B. stomach
- C. kidney
- D. large intestine

Q23. A lightning conductor protects a building by providing:

- A. a safe conducting path to Earth
- B. more friction
- C. a source of lightning
- D. an insulating cloud

Q24. A force of 50 N acts on area 0.5 m². Pressure is:

- A. 25 Pa
- B. 50 Pa
- C. 100 Pa
- D. 250 Pa

Q25. Two students are caught outside during a thunderstorm student. A stand under a large tree, while student B moves away from the tree and crouches low in a safer location until shelter is available. Which action is safer?

- A. Student A, because trees provide protection
- B. Student B, because tall is dated objects can be struck by lighting
- C. Both are equally safe
- D. Neither student is affected by lightning

Q26. A Student shines a torch on a plane mirror at an angle of 30° with the normal. At what angle will the reflected ray make with the normal?

- A. 15°
- B. 30°
- C. 60°
- D. 90°

Q27. A jeweler wants to give a silver appearance to a cheaper metal ornament. Which process can be used?

- A. Filtration
- B. Electroplating
- C. Evaporation
- D. Sedimentation

Q28. Increasing greenhouse gases tends to:

- A. increase global warming
- B. stop all sunlight
- C. freeze all oceans immediately
- D. remove the atmosphere

Q29. A farmer removes trees from a will side to create agricultural land. During heavy rain the top soil is washed away. What is the main reason?

- A. Trees increase soil erosion
- B. Trees roots keep hold the soil together
- C. Trees prevent water from entering soil
- D. Soil becomes heavier after deforestation.

Q30. Crop rotation helps maintain soil fertility by:

- A. varying nutrient demands and including useful crops such as legumes
- B. removing all microbes
- C. using the same crop forever
- D. preventing all water entry

SECTION C – ENGINEERING, TECHNOLOGY & DATA

Q31. A hydraulic lift works mainly because pressure in an enclosed fluid is:

- A. transmitted throughout the fluid
- B. lost completely
- C. converted only to heat
- D. independent of force

Q32. A 12 V supply across a 6 ohm resistor gives current:

- A. 0.5 A
- B. 2 A
- C. 6 A
- D. 72 A

Q33. A machine receives 800 J and gives 600 J useful output.

Q36. Which graph best compares categories such as crop yield of four fields?

- A. Bar graph
- B. Line graph only
- C. Scatter plot only
- D. Number line

Q37. A model bridge fails at its joints. Which redesign directly targets the weakness?

- A. Reinforce the joints
- B. Make spans longer
- C. Add weight at the centre
- D. Remove cross-bracing

Efficiency is:

- A. 60%
- B. 70%
- C. 75%
- D. 80%

Q34. A vehicle travels 72 km/h. This equals:

- A. 10 m/s
- B. 15 m/s
- C. 20 m/s
- D. 25 m/s

Q35. A wind turbine's blade length is doubled while shape and wind remain similar. The swept area becomes approximately:

- A. 2 times
- B. 3 times
- C. 4 times
- D. 8 times

Q38. A robot moves 8 m East, 6 m North, 3 m West, 2 m South. Its final coordinates relative to start are:

- A. 5 m East, 4 m North
- B. 11 m East, 8 m North
- C. 5 m West, 4 m North
- D. 8 m East, 2 m North

Q39. A student tests whether salt affects boiling point. The controlled variable should include:

- A. same amount of water and heating setup
- B. different amounts of water
- C. different thermometers each time
- D. different heating powers

Q40. A solar panel is rated 200 W and receives full Sun for 5 h. Ideal energy is:

- A. 0.2 kWh
- B. 0.5 kWh
- C. 1.0 kWh
- D. 5 kWh

SECTION D – STEM OLYMPIAD CHALLENGE

Q41. A 3:1 gear reduction means the output shaft typically turns:

- A. slower with greater torque
- B. faster with less torque always
- C. at identical speed
- D. without torque

Q42. Which material is best for a heat-insulating container?

- A. Foam
- B. Copper
- C. Aluminium
- D. Iron

Q43. A child accidentally gets an unknown liquid in her hand. Before touching it further, she wants to determine whether it is acidic or basic. Which method is appropriate for a school laboratory?

- A. Taste the liquid
- B. Smell it directly
- C. Test it by using a suitable indicator
- D. Touch it with bare hands

Q44. A 2 kg cart accelerates at 3 m/s². Net force is:

- A. 1.5 N
- B. 5 N
- C. 6 N
- D. 9 N

Q45. A dataset is 40, 42, 39, 41, 120. Which value is an outlier?

- A. 39
- B. 40
- C. 42
- D. 120

Q46. Which design process is most appropriate?

- A. Identify need - brainstorm - prototype - test - improve
- B. Build once - never test
- C. Copy blindly - decorate
- D. Test before identifying the problem

Q47. A convex lens can form a real image when the object is:

- A. beyond the focal point
- B. always at the lens centre only
- C. behind an opaque screen
- D. inside the mirror

Q48. An object floats in water when the upward buoyant force is:

- A. equal to its weight in equilibrium
- B. always zero
- C. less than its weight while stationary
- D. unrelated to weight

Q49. A greenhouse model traps heat mainly because transparent walls allow much incoming sunlight but reduce:

- A. loss of heat from inside
- B. gravity
- C. air pressure to zero
- D. all visible light

Q50. If a fair die is rolled twice, probability of getting 6 both times is:

- A. 1/6
- B. 1/12
- C. 1/36
- D. 1/72